

Yamin Mir

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OBJECTIVE

Mechanical Engineering student with a strong focus on CAD-based mechanical design, modeling, and analysis. Experienced in developing manufacturable components and assemblies supported by simulation, prototyping, and hands-on fabrication. Seeking engineering roles centered on mechanical design, product development, and applied CAD skills.

EDUCATION

- University of Illinois at Chicago** Chicago, IL
 - Bachelor of Science - Mechanical Engineering; Minor: Computer Science GPA: 3.5/4.0 Expected: December 2026*
 - Courses: Finite Element Analysis, Thermodynamics, Statics, Circuit Analysis, Engineering Design (ME250), Calculus I-III, Differential Equations, Python, C (CS107), MATLAB (CS109), Fluid Mechanics, Electricity & Magnetism, Intermediate Thermodynamics, Strength of Materials*

SKILLS SUMMARY

- Languages:** Python, C, MATLAB
- Frameworks:** ROS 2, OpenCV, Gazebo, MS Office
- CAD & Fabrication:** SolidWorks, Autodesk Inventor, 3D Printing, Laser Cutting, Finite Element Analysis (FEA)
- Hardware:** NVIDIA Jetson, Raspberry Pi, Arduino, VESC, Soldering, Machine Shop tools
- Robotics & Sensors:** URDF/Xacro, TF Trees, Odometry, Intel RealSense, LiDAR, IMU Integration

EXPERIENCE

- Research Assistant 2 - Robotics and Motion Lab** Chicago, IL
 - University of Illinois at Chicago* Aug 2024 - Present
 - Designed and simulated robots using SolidWorks for academic and lab-based projects
 - Utilized 3D printing and laser cutting to fabricate functional robot components
 - Presented research findings at weekly lab meetings and contributed to outreach events
- Mechanical Design Intern** Chicago, IL
 - University of Illinois at Chicago - GPIP Program* Jun 2024 - Aug 2024
 - Modeled robotic car for laser cutting and optimized components for efficient assembly
 - Programmed Raspberry Pi and Arduino to control motors and servo-based robotic arm
 - Developed real-time object detection system using Python, OpenCV, and ArUco markers

PROJECTS

- Pentagon Robot - Five-Bar Parallel Planar Manipulator** Fall 2025 - May 2026
 - Senior Design Project (ME-396)*
 - Leading design of automated pick-and-place system for transferring ping-pong balls using five-bar linkage mechanism
 - Performing inverse kinematics analysis to achieve $\pm 0.5\text{mm}$ accuracy and $\leq 5\text{s}$ cycle time
 - Designing mechanical assembly in SolidWorks with motion simulation verification
 - Implementing hybrid control using Arduino UNO R3 and Raspberry Pi for trajectory planning
 - Integrating vacuum end-effector with pressure sensor feedback for ball pickup
- Autonomous Racing Car Development** Jun 2025 - Present
 - Undergraduate Research Project*
 - Building 1/10-scale autonomous vehicle using NVIDIA Jetson Orin Nano and ROS 2 Humble
 - Integrated BLDC motor with VESC controller for precise speed/torque control via UART
 - Modeled car structure using URDF/Xacro with accurate inertial properties for Gazebo
 - Implementing Nav2 stack for autonomous navigation with planned RL integration
- Internal Flow Analysis of Non-Uniform Channel** Fall 2025
 - ME 211 - Fluid Mechanics Project*
 - Performed CFD analysis of incompressible internal flow through a 1 m non-uniform channel using ANSYS Fluent
 - Developed SolidWorks geometry and generated high-resolution mesh (900k elements) with refined boundary layers
 - Implemented laminar and turbulent ($k-\omega$ SST) models using mass flow rate inlet boundary conditions
 - Analyzed velocity profiles, pressure drop, Reynolds number, and wall shear stress for flow characterization
 - Validated numerical results against theoretical relations and discussed deviations due to turbulence modeling and geometry
- Autonomous Pinball Machine** Spring 2024
 - ME250 Design Project*
 - Designed and manufactured pinball machine maintaining 1/2" ball in perpetual motion
 - Created detailed CAD models and performed mechanism simulations using SolidWorks
 - Fabricated custom gears using 3D printing and integrated Arduino-based sensors